Calcurion Maths Worksheets

A-Level Mathematics

Topic: Statistics – Normal Distribution

Medium & Hard • 25 Questions

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Section A – Medium (Q1–15)

1. $X \sim N(50, 16)$. Find $P(X < 54)$. **[3]**

2. $X \sim N(100, 25)$. Find $P(X > 108)$. **[3]**

3. $X \sim N(20, 9)$. Find $P(17 < X < 23)$. **[4]**

4. $Z \sim N(0, 1)$. Find $P(Z < 1.5)$. **[2]**

5. $Z \sim N(0, 1)$. Find $P(|Z| < 1.96)$. **[3]**

6. $X \sim N(\mu, 4)$. Given $P(X < 10) = 0.3$, find μ. **[4]**

7. $X \sim N(30, \sigma^2)$. Given $P(X > 35) = 0.2$, find σ. **[4]**

8. State the mean and variance of the standard normal distribution. **[2]**

9. $X \sim N(60, 36)$. Find the value of a such that $P(X < a) = 0.9$. **[4]**

10. $X \sim N(25, 16)$. Standardise $x = 29$ and find $P(X > 29)$. **[3]**

11. Heights of plants are $N(15, 4)$. Find the probability a plant is taller than 18 cm. **[3]**

12. $X \sim N(50, 100)$. Find $P(40 < X < 55)$. **[4]**

13. Sketch the normal curve $N(0, 1)$ and shade $P(Z > 1.2)$. **[2]**

14. $X \sim N(\mu, \sigma^2)$. Write $P(X > a)$ in terms of Φ. **[2]**

15. Approximately 68% of data from a normal distribution lie within how many SD of the mean? **[1]**

Section B – Hard (Q16–25)

16. $X \sim N(100, 64)$. Find c such that $P(|X - 100| < c) = 0.95$. **[5]**

17. Scores are $N(70, 49)$. The top 10% receive a distinction. Find the minimum score for a distinction. **[4]**

18. $X \sim N(\mu, 25)$. Given $P(X < 40) = 0.1$ and $P(X > 55) = 0.05$, find μ and check consistency. **[6]**

19. Lifetimes of bulbs are $N(800, 100^2)$. Find the probability a bulb lasts between 750 and 900 hours. **[4]**

20. $X \sim N(50, 16)$. Find the interquartile range of X. **[5]**

21. A continuous random variable has pdf that is normal. Explain why $P(X = x) = 0$ for any specific x. **[2]**

22. $X \sim N(20, 9)$, $Y \sim N(15, 4)$ independent. Find $P(X + Y > 40)$. **[5]**

23. Using inverse normal, find z such that $\Phi(z) = 0.975$. **[2]**

24. Weights of apples are $N(120, 25)$. Packs of 4 are made. Assuming independence, find the distribution of the total weight and $P(\text{total} > 490)$. **[6]**

25. Explain the continuity correction when approximating a discrete distribution by a normal. **[3]**

End of Worksheet • *Show all working clearly. Use normal tables or calculator as appropriate.*